Emoticon-Aware Recurrent Neural Network Model for Chinese Sentiment Analysis

1st Da Li
Graduate School of Information Science and Technology
Hokkaido University
Sapporo, Japan
lida@ist.hokudai.ac.jp

2nd Rafal Rzepka
Graduate School of Information Science and Technology
Hokkaido University / RIKEN AIP
Sapporo, Japan
rzepka@ist.hokudai.ac.jp

3rd Michal Ptaszynski
Department of Computer Science
Kitami Institute of Technology
Kitami, Japan
ptaszynski@cs.kitami-it.ac.jp

4th Kenji Araki
Graduate School of Information Science and Technology
Hokkaido University
Sapporo, Japan
araki@ist.hokudai.ac.jp

***Abstract*—Pictograms (emoticons/emojis) have been widely used in social media as a mean for graphical expression of emotions. People can express delicate nuances through textual information when supported with emoticons, and the effectiveness of computer-mediated communication (CMC) is also improved. Therefore it is important to fully understand the influence of emoticons on CMC. In this paper, we propose an emoticon polarity-aware recurrent neural network method for sentiment analysis of Weibo, a Chinese social media platform. In the first step, we analyzed the usage of 67 emoticons with facial expression used on Weibo. By performing a polarity annotation with a new "humorous type" added, we have confirmed that 23 emoticons can be considered more as humorous than positive or negative. On this basis, we applied the emoticons polarity in a Long Short-Term Memory recurrent neural network (LSTM) for sentiment analysis of undersized labelled data. Our experimental results show that the precision for predicting sentiment polarity on Weibo.**

***Index Terms*—emoticons, LSTM, social media, sentiment analysis**

I. Introduction

People have become increasingly accustomed to expressing their opinions online, especially on social media such as Twitter, Facebook or Weibo - the biggest Chinese social media network that was launched in 2009. This microblog data contain a vast amount of valuable sentiment information not only for the commercial use, but also for psychology, cognitive linguistics or political science. According to the study on Instagram, emojis are present in up to 40% of online messages in many countries [1]. For example, 😂 "Face with Tears of Joy", an emoji that means the person has an extremely good mood, was regarded as the 2015 word of the year by The Oxford Dictionary [2].

Sentiment analysis of microblogs becomes an important area of research in the field of Natural Language Processing. Study of sentiment in microblogs in English language has undergone major developments in recent years [3]. Chinese sentiment analysis research, on the other hand, is still at early stage [4] especially in the domains of lexicons and emoticons. In our opinion ignoring emoticons in sentiment research is unjustifable, because they convey a significant emotional information and play an important role in expressing emotions and opinions in social media [5], [6].

In this paper we focus on the emoticons used on Weibo; we analyze the characteristics of this particular set of emoticons, report on their evaluation while dividing them into three categories: positive, negative and humorous. We list 23 emoticons which were annotated as "humorous" more often than "positive" or "negative". We also noticed that among the resources of Chinese social media sentiment analysis, the labelled Weibo data containing emoticons are extremely rare which makes considering them in machine learning approaches difficult. To resolve this problem, we applied the emoticons polarity in a Long Short-Term Memory recurrent neural network (LSTM) in a way which allows sentiment analysis on smaller annotated data sets. Our experimental results show that the proposed method can significantly improve the performance of sentiment analysis on Weibo.

Our main contributions are as follows:

- We empirically confirmed implicit humor characteristic to Chinese culture visible on Weibo and divided Weibo emoticons into three categories: positive, negative and humorous.
- We applied the emoticons polarity in a Long Short-Term Memory recurrent neural network (LSTM) proposing an emoticon polarity-aware Long Short-Term Memory (EP-LSTM) method for small labelled data to be used in sentiment analysis (see Figure 1 for the overview of our method).

II. Related Research

At present, the sentiment analysis technology generally can be divided into two categories: rule-based methods relying

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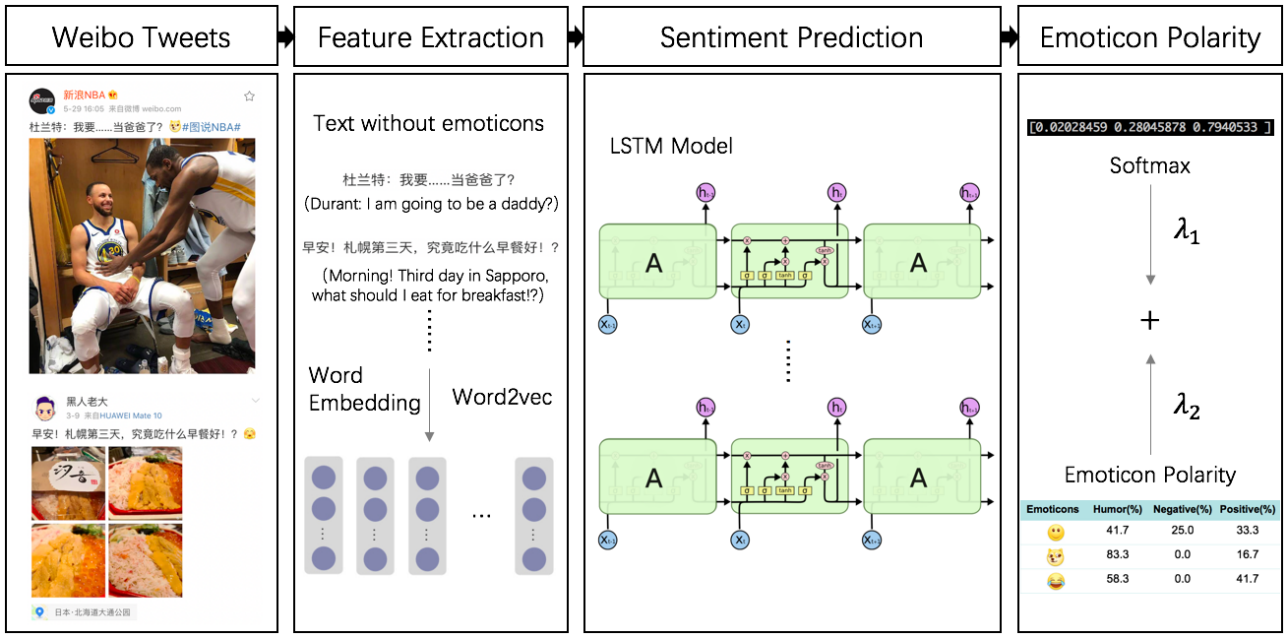


Fig. 1. Overview of the proposed EP-LSTM Method.

on sentiment lexicons, and machine learning-based methods relying on annotated data.

A. Rule-based methods

Zhang et al. [7] proposed a rule-based approach with two phases: a) the sentiment of each sentence is first decided based on word dependency to aggregate the sentences sentiments and then b) the sentiment of each document is calculated. Zagibalov et al. [8] presented a method that does not require any annotated corpus training data and only requires information on commonly occurring negations and adverbials. Li et al. [9] stated that polarities and strengths judgment of sentiment words comply with a Gaussian distribution, and thus proposed a Normal distribution-based sentiment computation method which allows quantitative analysis of semantic fuzziness of sentiment words in Chinese language. Zhuo et al. [10] presented a novel approach based on the fuzzy semantic model by using an emotion degree lexicon and a fuzzy semantic model. Their model includes text preprocessing, syntactic analysis, and emotion word processing. However, optimal results of Zhuo's model were achieved only when the task was clearly defined.

B. Machine learning-based methods

Tan and Zhang [11] conducted an empirical study of sentiment categorization on Chinese documents. They tested four features – mutual information, information gain, chi-square, and document frequency; and five learning algorithms: centroid classifier, k-nearest neighbor, winnow classifier, Naïve Bayes (NB) and support vector machine (SVM). Their results showed that the information gain and SVM features provided the best performances for sentiment categorization coupled with domain or topic dependent classifiers. There are also researchers who have combined the machine learning approach with the lexicon-based approach. Chen et al. proposed a novel sentiment classification method which incorporated existing Chinese sentiment lexicon and convolutional neural network [12]. The results showed that their approach outperforms the convolutional neural network (CNN) model only with word embedding features [13]. However, all these approaches did not consider emoticons.

Recently, a powerful system utilizing emoji in Twitter sentiment analysis model called DeepMoji was proposed [14]. Its creators trained 1,246 million tweets containing one of 64 common emoticons by Bi-directional Long Short-Term Memory (Bi-LSTM) model and applied it to interpret the meaning behind the online messages. DeepMoji is also the most advanced sarcasm-detecting model, with an accuracy rate of 82.4% even outperforming human detectors who managed to acquire 76.1% accuracy rate. Sarcasm reverses the emotion of the literal text, therefore sarcasm-detecting capability can play a significant role in sentiment analysis, especially in case of social media. Although sarcasm and irony tend to convey negative emotions in general, we found that in Chinese social media (Weibo in our example), in addition to the expression of positive and negative emotions, people tend to express a kind of humorous emotion that escapes the traditional bi-polarity.

III. Analysis of Emoticons in Chinese Social Media

In the real-life (offline) dialogue between human beings, besides tone changes, we usually express emotions with body language. In social networks, this can partially be achieved by using emoticons [15].

There are many unknown factors in constantly changing moods of human beings, but communication with emoticons

has become a global phenomenon. On the other hand, because of different ethnic and cultural differences, misunderstandings when using facial emoticons is not uncommon [16]. We have noticed previously mentioned humorous emotion in Weibo microblog entries containing emoticons which are often difficult to interpret as positive or negative. It seems that some emoticons are used just for fun, self-mockery or jocosity which expresses an implicit humor characteristic in Chinese culture. Emoticons seem to play an important role in expressing this kind of emotion. There is a high possibility that this phenomenon can cause a significant difficulty in sentiment detecting task, therefore we decided to analyze Weibo emoticons before adding them to our system for classifying emotions in Chinese microblogs.

For the analysis we chose 67 most common emoticons out of 144 emoticons used on Weibo (see Figure 2) which represent facial expression, because this type is usually used in sentiment analysis in other languages. We asked 12 Chinese native speakers to label these 67 emoticons by applying one of three following categories: positive, negative and humorous. The evaluators were 6 males and 6 females in their 20's, each of them being a Weibo user for more than five years.

The annotation results showed that there are 44 emoticons with obvious positive or negative emotion, and the remaining 23 emoticons are considered more as humorous than positive or negative. The results are summarized in Table I.

Our hypothesis is that the emoticons which are considered to be humorous in more than 50% of cases (see Table I) usually convey tongue-in-cheek type of message. Moreover, the sentiment polarity of emoticons such as , , and other cases show rather clearly their ambiguity, and they should be treated with special care when we add emoticon recognition capability to our system in the future. was originally a smile emoji, but it seems that gradually Weibo users started using this emoji for expressing artificial smile, ridicule or self-mockery. On the other hand, when dealing with such emoticons on Weibo, we should be careful with jumping to conclusion and straightforwardly label it as a humorous. Instead we need to use a significant number of labelled examples containing these emoticons to train a machine learning model.

Therefore, we designed a deep learning network with considering emoticons to predict emotions in Weibo microblogs in a more sophisticated manner. Details are introduced in the following section.

TABLE I
TWENTY-THREE EMOTICONS CONVEYING HUMOR TYPICAL FOR CHINESE CULTURE.

Emoticon	Humorous (%)	Negative (%)	Positive (%)
	41.7	25.0	33.3
	58.3	0.0	41.7
	66.7	33.3	0.0
	91.7	8.3	0.0
	58.3	0.0	41.7
	83.3	0.0	16.7
	58.3	25.0	16.7
	66.7	8.3	25.0
	66.7	8.3	25.0
	41.7	33.3	25.0
	75.0	25.0	0.0
	58.3	41.7	0.0
	50.0	25.0	25.0
	66.7	33.3	0.0
	50.0	50.0	0.0
	58.3	41.7	0.0
	50.0	50.0	0.0
	50.0	50.0	0.0
	50.0	33.3	16.7
	41.7	41.7	16.7
	75.0	8.3	16.7
	58.3	33.3	8.3
	75.0	0.0	25.0

IV. PROPOSED METHOD

Inspired by above mentioned observations, we propose a emoticon polarity-aware Long Short-Term Memory (LSTM) model for sentiment classification on Weibo. We use LSTM [17] as the framework because it can overcome the problem of gradient vanishing compared with recurrent neural network (RNN). An overview of the proposed method for emoji prediction is shown in Figure 1. In the first step, we extract the textual features from a Weibo post containing only one type of emoticon. Then, the textual features are sent into the LSTM model to learn an output representation. Second, we use a softmax classifier to obtain the predicted results of textual features and output their probability. Next, we give a hyperparameter for each text and emoticon, calculate the summation of both features with the hyperparameter. Finally, we can obtain the sentiment probability of a Weibo post which considers the effect of emoticons.

A. Model structure

We explain our model in Figure 3. An LSTM network computes a mapping from an input sequence $x = (x_1, ..., x_T)$ to an output sequence $y = (y_1, ..., y_T)$ by calculating the network unit activations using the following equations iteratively from $t = 1$ to T. The equations of the LSTM cell are as follows:

$$i_t = \sigma(W_{ix}x_t + W_{im}m_{t-1} + W_{ic}c_{t-1} + b_1) \quad (1)$$

$$f_t = \sigma(W_{fx}x_t + W_{fm}m_{t-1} + W_{fc}c_{t-1} + b_f) \quad (2)$$

$$c_t = f_t \odot c_{t-1} + i_t \odot g(W_{cx}x_t + W_{cm}m_{t-1} + b_c) \quad (3)$$

$$o_t = \sigma(W_{ox}x_t + W_{om}m_{t-1} + W_{oc}c_t + b_o) \quad (4)$$

$$m_t = o_t \odot h(c_t) \quad (5)$$

$$y_t = \phi(W_{ym}m_t + b_y) \quad (6)$$



Fig. 2. Most common emoticons of Weibo.

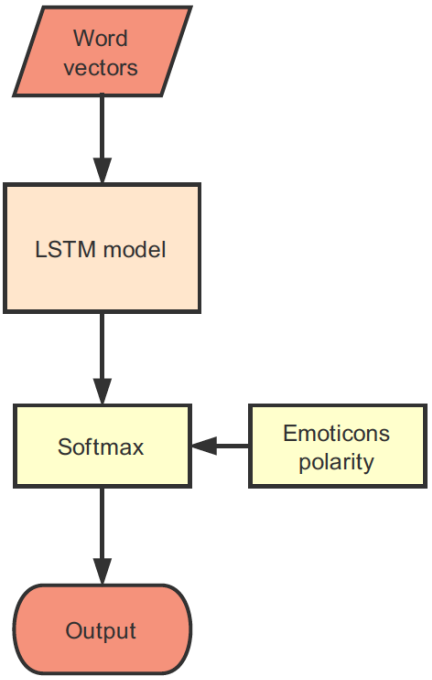


Fig. 3. Flowchart of the proposed EP-LSTM method for sentiment analysis.

where the W terms denote weight matrices (e.g. W_{ix} is the matrix of weights from the input gate to the input), W_{ic}, W_{fc}, W_{oc} are diagonal weight matrices for peephole connections, the b terms denote bias vectors (b_i is the input gate bias vector), σ is the logistic sigmoid function, and i, f, o and c are respectively the input gate, forget gate, output gate and cell activation vectors, all of which are the same size as the cell output activation vector m, $\odot$ is the element-wise product of the vectors, g and h are the cell input and cell output activation functions, generally and also in this paper $tanh$, and ϕ is the network output activation function, softmax in our method.

The outputs of softmax layer $S(z_i)$ are the probability of each category. The softmax function is defined as following [18]:

$$S(z_i) = \frac{e^{z_i}}{\sum_{j=1}^{i} e^{z_j}} \tag{7}$$

where the input of softmax layer z_i is defined as:

$$z_i = w^i x + b_i \tag{8}$$

where w is the weight and b is bias, both of them are calculated by training the model.

B. Emoticon polarity

In order to predict sentiment category of Weibo posts considering the influence of emoticons for Chinese social media sentiment analysis, we give the probability of the LSTM model's softmax output $S(z_i)$ a hyperparameter λ_1. At the same time, we take the labelled emoticons as polarity P_e, and assign a hyperparameter λ_2. P becomes the final probability output of the classification:

$$P = \lambda_1 S(z_i) + \lambda_2 P_e \tag{9}$$

where the summation of λ_1 and λ_2 is equal to 1.

V. Experiments

In order to verify the validity of our proposed method, we made the following experiments.

A. Preprocessing

Initializing word vectors with those obtained from an unsupervised neural language model is a popular method to improve performance in the absence of a large supervised training set. Considering that there are many neologisms and Internet slang expressions on Weibo, for our experiment we collected a large dataset (7.31 million posts) from Weibo API from May 2015 to July 2017 to be used for word embedding. First, we deleted the emoticons, images, and videos treating them as noise. Next, we used Python Chinese word segmentation module Jieba[1] to segment the sentences

[1]https://github.com/fxsjy/jieba

TABLE II
PRECISION COMPARISON RESULTS.

	Humorous	Negative	Positive
Emoticon Only Polarities	60.8%	0.0%	0.0%
LSTM	58.7%	50.0%	26.9%
EP-LSTM	63.0%	75.0%	62.5%

of the microblogs, and applied the segmentation results into the Word2vec model [19] for training word vectors. The vectors have dimensionality of 100 and were trained using the continuous skip-gram architecture.

After that, we collected 3,000 Weibo posts containing the 🙂, 😺 , 😂, 🐥 emoticons, ensuring each entry has only one emoticon of given type (cases with more emoticons of the same type were allowed). To use these posts as our training data, we asked three Chinese native speakers to annotate them into three categories: "positive", "negative", and "humorous". After one annotator labelled polarities of all posts, two other native speakers confirmed correctness of his annotations. Whenever there was a disagreement, all decided the final polarity through discussion.

B. LSTM model

We deleted all emoticons from training data. After training a word embedding model, we passed its output into the LSTM model to train and save the model. We trained the model with 10 epochs and the performance achieved the highest value when the dropout rate was 0.5 and the batch size was 32. The validity of the model was examined by holdout method (95%/5%, training/validation) and it achieved 78.0% accuracy.

C. Testing

We collected 120 Weibo entries with the four emoticons mentioned above, deleted the emoticons and processed sentiment classification with the trained LSTM model using the softmax layer which outputs probability of each of three categories. Then we used the proposed method (described in Section 4) to calculate probability of each category and confirmed the precision. Because we assumed that in emotion expression emoticons might play a greater role than text, in our experiment, we set the hyper-parameters λ_1 and λ_2 to 0.4 and 0.6 respectively. As the baseline methods, we compared the results of sentiment classification by emoticon polarity and of LSTM network only. The results are shown in Table II.

The results showed that our proposed method is more effective than a) LSTM network only categorization and b) considering just emoticon polarities. Limited to small annotated data, the sentiment classification precision predicted by the LSTM model is relatively low, but our proposed approach has improved the performance showing that low-cost, small-scale data labelling is able to outperform widely used state-of-the-art methods.

VI. CONSIDERATIONS

In our proposed EP-LSTM approach, we paid more attention to the weight of the emoticons in microblogs, therefore compared to LSTM, the proposed method obtained better performance in some unclear posts. For example (see Figure 4), this is a microblog of correct prediction with our EP-LSTM method, which is labeled as "humorous", but it is predicted by LSTM as "negative". There is an idiom in Chinese culture: "I swear if I tell half a lie, I will be killed by the thunder and God". So the microblog expressed a humorous view of the natural phenomenon, and we found that our proposed method can predict the implicit humorous meaning to correct category with considering the polarity of the emoticons.

昨天美国堪萨斯州出现了奇怪的闪电，是从下往上的，是因为发誓的人太多了吗？😺

Yesterday, there was a strange thunder in Kansas, USA. It was from the ground up to the sky. Is it because there were so many people swearing (they have not lied)? 😺

Fig. 4. Example of correct prediction.

We also analyzed the wrongly predicted examples of results. We found that the short text or text with images brings an impact on the results. Weibo microblogs contain a lot of wrongly chosen characters and new words which are difficult to detect, especially for a short entry containing only a few words. We think that adding image processing or adding a lexicon of neologisms may improve the results in the future.

An example of short text with unconventional character choice and images of Weibo microblog is shown in Figure 5 below.



Fig. 5. Example of unconventional Chinese character choice in a Weibo post with images. Entry says "A notch on the screen.... speechless.", however one of characters of the word "notch" is replaced with another homonymous one changing the meaning to person's name.

We also found that some Weibo entries posted with a humorous expression like self-mockery also contain optimistic or pessimistic emotional load. It is quite easy to make a wrong prediction by simply classifying them into "humorous" category. Therefore, A fine-grained humorous classification should be considered in the future.

An example of Weibo entry predicted as "humorous" and annotated as "negative" is shown in Figure 6 below.

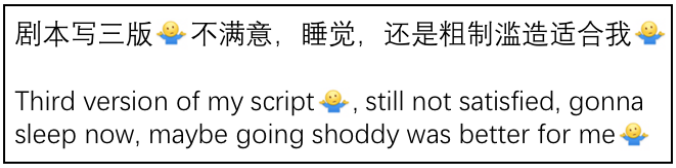


Fig. 6. Example of wrong classification into "humorous" category.

VII. Conclusions and Future Work

In this paper, we analyzed the characteristics of a particular set of emoticons: "humorous", and reported its human-based evaluation after dividing into three categories: positive, negative and humorous. We listed 23 emoticons which were annotated as "humorous" more often than "positive" or "negative". We applied the emoticons polarity in a Long Short-Term Memory recurrent neural network (LSTM) and proposed an Emoticon Polarity-aware Long Short-Term Memory (EP-LSTM) method for sentiment analysis for small labelled datasets. Our experimental results show that the proposed method can significantly improve the precision for predicting sentiment polarity on Weibo.

For improving the performance of the proposed method, in near future we are going to increase the amount of labelled data, so that we can expect to acquire the hyperparameters automatically by machine learning approaches. Furthermore, we need to increase the number of evaluators for annotating Weibo emoticons in fine-grained humorous emotion to enhance the reliability of the polarity of emoticons.

In order to achieve effective Chinese sentiment analysis method, we will also consider the semi-supervised learning and other approaches to train our model, meanwhile, we will consider a method for automatic annotation for Weibo data with emoticons with machine learning.

In addition, we also confirmed that there is a significant number of neologisms and online jargon in social media that contain rich emotions. In the future work, we will expand the Chinese sentiment lexicon, and then by combining the enriched polarity of emoticons, deep learning and sentiment lexicons we will construct a high-quality sentiment recognizer which deals with wider spectrum of sentiment in Chinese language.

VIII. Acknowledgment

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